

Course	Surface Water Engineering	
		Hydraulic Structure Design
Topic	Chute Design	
Version	2024	

This handout gives you a step by step procedure to design a standard 33° chute. The standard chute is illustrated in Figure 1 and hydraulic design information is provided in Figure 2.

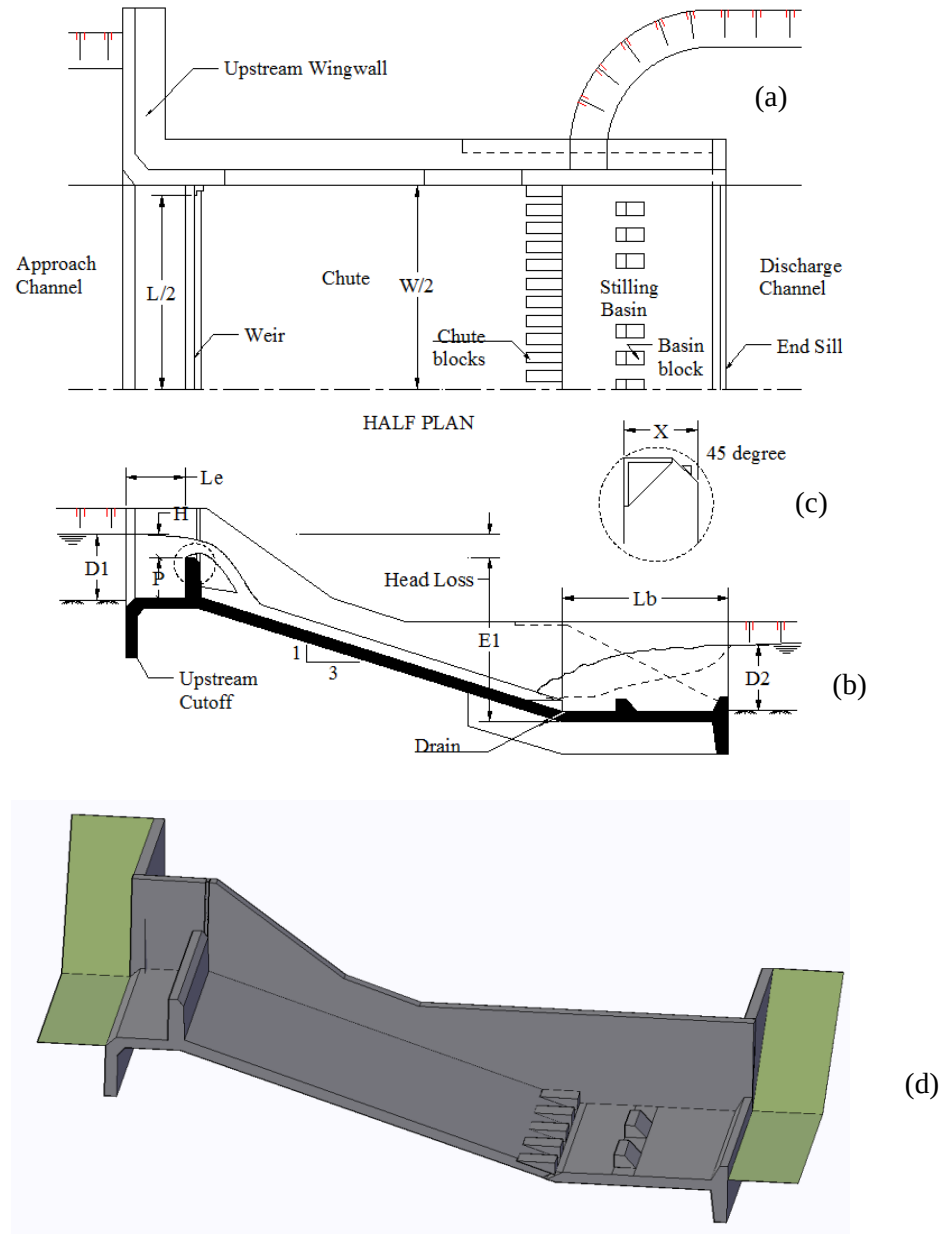


Figure 1. Definition sketch for a Chute Drop Structure (a) horizontal half plan plane; (b) profile plane; (c) weir with steel protection angle and thickness X ; and (d) 3D sketch

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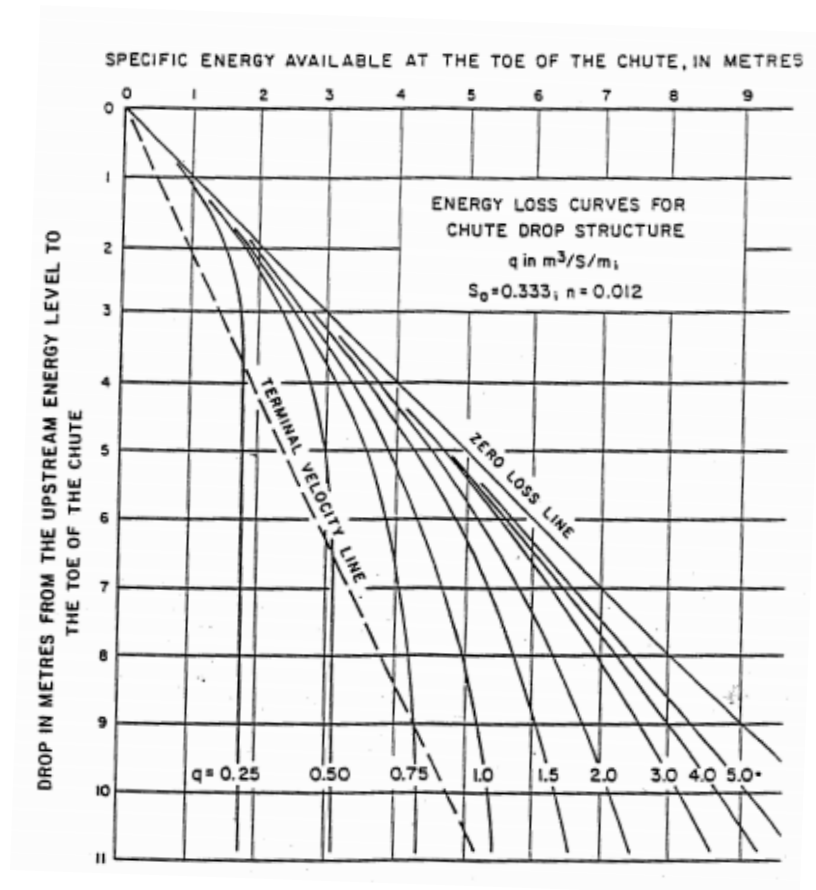


Figure 2. Energy loss curve for chute drop structure design.

Main steps to design a chute:

1. Determine maximum expected discharge Q through chute.
2. Width (W) of the chute must be large enough to permit water discharge and it is usually smaller than the actual channel width. The following relationship is given which relates the required width (W) to the discharge flow (Q):

$$W = 2.3 \sqrt{Q}$$

3. The weir height (P) is determined by the following equations:

$$P = D_1 + \frac{V_1^2}{2g} - H$$

Where V_1 is flow velocity in channel

H is total head on weir

D_1 is the normal depth in the channel

$$Q = 1.837 (L) H^{1.5} \text{ (weir equation for a weir of length } L \text{)}$$

The weir thickness (X), as shown in Figure 1c, is found by using the following relationship:

$$X = 0.13 H$$

The weir edge can be protected from damage by having an angle iron imbedded in the concrete as shown in Figure 1c.

4. Weir nappe ventilation design: To avoid a clinging nappe, which could lead to potential damage of the chute, two common methods for nappe ventilation are recommended:

- Installing properly sized pipes in the walls to vent the cavity under the nappe as shown in Figure 3b.
- Use the corner of an angle iron pointed upstream to slightly constrict the flow and allow air under the nappe (Figure 3a). The iron angle width (Y) can be calculated using this relationship:

$$Y = \sqrt{0.0008 \cdot W \cdot H}$$

The length of the weir (L) then becomes:

$$L = W - 2Y$$

The weir equation then becomes:

$$Q = 1.837 (W - 2Y) H^{1.5}$$

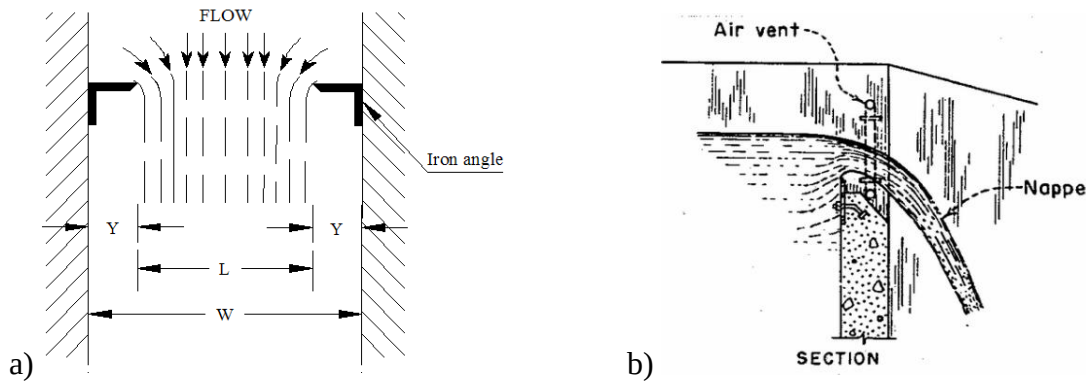


Figure 3. a) Nappe ventilation by pipes imbedded in side walls and b) ventilation using steel/iron angles on the side walls.

5. Stilling Basin Design:

Energy at the bottom of the chute will be dissipated by a hydraulic jump. The length of the stilling basin is related to the height of the jump or the natural depth of water in the downstream channel:

$$L_b = 3 \cdot y_2 \text{ or } 3 \cdot D_2$$

Where y_2 is the depth after the hydraulic jump (calculated as shown below); D_2 is natural downstream channel depth

$$y_2 = \frac{-y_1}{2} + \sqrt{\frac{2V_1^2 y_1}{g} + \frac{y_1^2}{4}}$$

Substituting the Froude number $F_1 = \frac{V_1}{\sqrt{gy_1}}$ results in $y_2 = (\sqrt{8F_1^2 + 1} - 1) \frac{y_1}{2}$

The depth (y_1) at the bottom of the chute is determined by solving the following two equations and by using the information provided in Figure 2 for the energy at the toe of the chute:

$$\left\{ \begin{array}{l} q = V_1 y_1 \\ \text{where unit design discharge } q = \frac{Q}{w} \\ E_1 = y_1 + \frac{V_1^2}{2g} \\ \text{Where } E_1 \text{ is the energy loss at toe of chute which depends on the chute slope,} \\ \text{drop height and unit design discharge} \end{array} \right.$$

6. To finalize the design, chute blocks, basin blocks (baffle blocks) and end sill are needed. The standard design of a chute uses a type III stilling basin as shown in Figure 4.

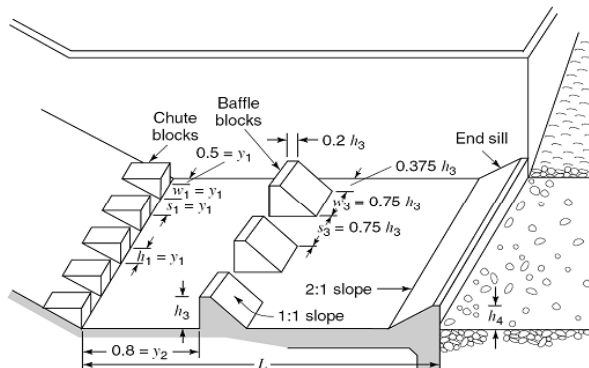


Figure 4. Basin Type III – For canal structures, small outlet works, and small spillways (U.S. Bureau of Reclamation, 1987).

As noted in Figure 4:

Chute blocks

$$\text{Height } h_1 = y_1 \text{ or } \frac{y_2}{q}$$

$$\text{Width } w_1 = \text{spacing} = y_1$$

Basin blocks (baffle blocks)

$$\text{Height } h_3 = 1.5 y_1 \text{ or } \frac{y_2}{8}$$

$$\text{Width } w_3 = 0.75 h_3$$

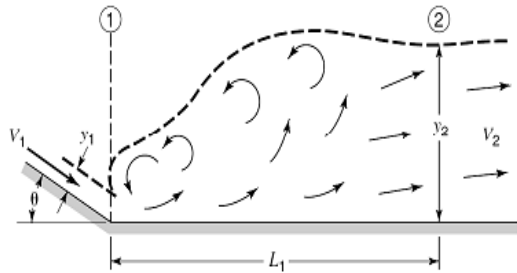
$$\text{Spacing } s_3 = 0.75 h_3$$

End sill

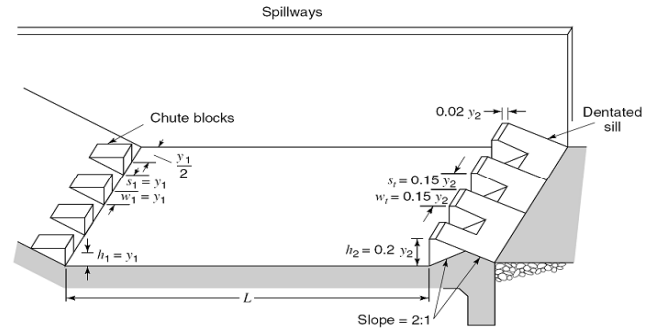
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$$\text{Height } h_4 = \frac{y_2}{10}$$

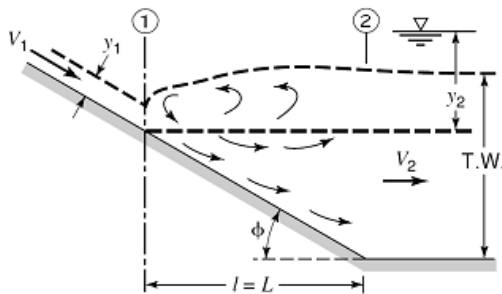
A variety of other stilling basins can also be chosen under different conditions.



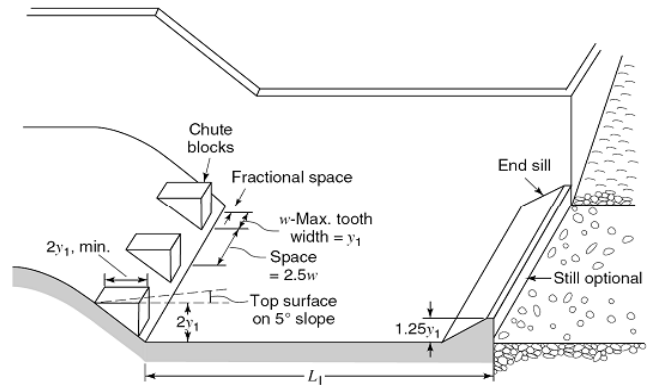
Basin Type I – Horizontal apron (U.S. Bureau of Reclamation, 1987)



Basin Type II – For high dam, earth dam spillways and large canal structures (U.S. Bureau of Reclamation, 1987)



Basin Type V – Sloping apron (U.S. Bureau of Reclamation, 1987).



Basin Type IV – For canal structures, outlet works, and diversion dams (U.S. Bureau of Reclamation, 1987)

Figure 5. Stilling basin options

Example

Design a chute to handle a $1.4 \text{ m}^3/\text{s}$ design discharge. The chute should have a 0.333 slope. . Manning's $n = 0.0122$ (concrete chute). The chute should have a 5 m drop.

Solution:

$$W = 2.3 \sqrt{Q} \rightarrow W = 2.72 \text{ m.}$$

$$q = 1.4 \text{ m}^3/\text{s} / 2.72 = 0.5 \text{ m}^3/\text{s}/\text{m}$$

$$E_1 = 3 \text{ m (at toe of chute) (Figure 2)}$$

$$\begin{cases} q = V_1 y_1 \\ E_1 = y_1 + \frac{v_1^2}{2g} \end{cases}$$

$$\begin{cases} 0.5 \frac{m^3}{sm} = V_1 y_1 \\ 3 m = y_1 + \frac{v_1^2}{2g} \end{cases} \rightarrow \begin{cases} V_1 = 7.6 m/s \\ y_1 = 0.066 m \end{cases} \quad \text{or} \quad \begin{cases} V_1 = 7.9 m/s \\ y_1 = 0.064 m \end{cases}$$

Note: How to solve a *General cubic equation*?

$$x^3 + bx^2 + cx + d = 0$$

Replace $p = c - \frac{b^2}{3}$ and $q = d + \frac{2b^3}{27} - \frac{bc}{3}$ result in

$$x^3 + px + q = 0$$

$$\Delta = q^2 + \frac{4p^3}{27}$$

If $\Delta > 0$:

$$x = \sqrt[3]{\frac{-q + \sqrt{\Delta}}{2}} + \sqrt[3]{\frac{-q - \sqrt{\Delta}}{2}}$$

If $\Delta < 0$:

$$x_1 = \sqrt[3]{\frac{-4p}{3}} \cos\left(\frac{G}{3}\right)$$

$$x_2 = \sqrt[3]{\frac{-4p}{3}} \cos\left(\frac{G+2\pi}{3}\right)$$

$$x_3 = \sqrt[3]{\frac{-4p}{3}} \cos\left(\frac{G+4\pi}{3}\right)$$

$$G = \arccos((3q\sqrt[3]{3})/(p\sqrt{-4p}))$$

If $\Delta = 0$:

$$x_1 = 3\frac{q}{p}$$

$$x_2 = x_3 = \frac{-3q}{2p}$$

Choose $\begin{cases} V_1 = 7.9 m/s \\ y_1 = 0.064 m \end{cases}$ (more critical)

$$y_2 = \frac{-y_1}{2} + \sqrt{\frac{2V_1^2 y_1}{g} + \frac{y_1^2}{4}} = \frac{-0.064}{2} + \sqrt{\frac{2 \cdot 7.9^2 \cdot 0.064}{9.81} + \frac{0.064^2}{4}} = 0.87 m$$

Length of the stilling basin $L_b = 3 y_2 = 3 \cdot 0.87 = 2.61 m$.

Choose Basin Type III for small canal structure

Chute blocks: $h_1 = y_1 = 0.07 m$ (height)

$$w_1 = y_1 = 0.07 \text{ m (width)}$$

$$s_1 = y_1 = 0.07 \text{ m (space)}$$

Baffle blocks: Slope 1:1

$$h_3 = 1.5 y_1 = 0.11 \text{ m}$$

$$w_3 = 0.75 h_3 = 0.08 \text{ m}$$

$$s_3 = 0.75 h_3 = 0.08 \text{ m}$$

End sill

$$h_4 = \frac{y_2}{10} = 0.09 \text{ m (height)}$$